

Abstract No.

Biomimetic Fluidic Model of the Glioblastoma-Affected Neurovascular Unit as an Advanced In Vitro Platform

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Glioblastoma (GBM) remains a poorly understood and highly aggressive brain tumor, characterized by frequent recurrence and limited therapeutic options. A major challenge in GBM treatment is the inability of many potentially effective drugs to cross the blood-brain barrier (BBB), the selective interface that protects the central nervous system.

Current in vitro BBB models are commonly based on static Transwell systems, which fail to reproduce the physiological shear stress generated by blood flow. As a result, drug transport and uptake dynamics often differ from those observed in vivo.

Here, we present a biomimetic in vitro platform designed to model a GBM-affected neurovascular unit (NVU) using exclusively primary human cells. Primary cells were isolated from GBM surgical specimens, phenotypically characterized, and separated into distinct cellular populations. In parallel, patient-derived GBM tumoroids were generated.

The resulting cell populations were seeded into a custom-designed perfused bioreactor according to the histological organization of the human NVU. The device consists of a three-dimensional dual-compartment structure. The upper channel reproduces the BBB and contains endothelial cells, astrocytes, and pericytes. The lower compartment, directly connected to the upper channel, contains GBM tumoroids together with healthy microglia, astrocytes, and neurons, mimicking the brain parenchymal environment. The bioreactor was engineered through fluid-dynamic modeling to generate a physiological wall shear stress of approximately 15 dyn/cm², comparable to that of human cerebral capillaries.

This platform provides a physiologically relevant tool for investigating BBB permeability, assessing the distribution of therapeutic agents within brain tissue, and evaluating treatment responses in GBM and surrounding neural cells. The model may support the development and preclinical testing of novel therapies for glioblastoma.



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